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MGM Macau and Cotai

Resource Efficiency Management Report

April 2021

Energenz Consulting Limited

EXECUTIVE SUMMARY

Energenz has been working closely with MGM to implement the EMOs as well as identify additional opportunities to achieve ISO certification energy targets. Energenz appreciates the great support and trust from MGM team over the years, in particular the Facility Management and the Sustainability Teams.

Key Performance Indicators

Energenz reports both normalized and actual savings on a monthly basis for both properties. The normalized savings are determined based on a 2019 baseline, with aligned annual savings targets to meet both ISO50001 as well as Energenz's retainer contract requirements. Please note that the reported normalized savings achieved are to-date and cumulative for each month of the respective year.

Table 1. ISO and Retainer Contract Energy Performance Targets for 2025

MGM Properties	Baseline	Energy Savings	Normalized Savings Achieved Up to April 2021	2022	2023	2024	2025
Macau	2019	Target	-4%	-8%	-12%	N/A	**-13%**
		Actual	**-9.6%**	-	-	-	**-**
Cotai	2019	Target	-4%	-8%	-12%	N/A	**-13%**
		Actual	**-11.5%**	-	-	-	**-**

**MGM Cotai ISO energy performance: e-bus consumption is included.*

The actual monthly savings are determined against the 2019 baseline consumption and cumulative for each month of 2021.

Table 2. Internal Actual Energy Performance Targets for 2025

MGM Properties	Baseline	Actual Saving Target[1]	Savings Achieved Up to April 2021
Macau	2019	15%	**-9.2%**
Cotai	2019	15%	**-10.0%**

Highlighted Changes in Operation

The LPG consumption in Macau has increased in the month of April as the building, starting March 2021, started to utilize the steam to water HX to raise DHW temperatures to disinfect the hot water system.

Similarly, Cotai's natural gas consumption has also increased to accommodate the disinfecting procedure. The temperature boost-up will be on-going until an alternative method is established. Currently, there is no end date identified.

[1] Actual Energy Savings Target for 2025 has been established based on future CAPEX projects, following review with MGM teams

Consumption Intensity

The YTD electricity and water use intensities are tracked on a monthly basis for both properties:

Table 3. Year to Date Actual Electricity Use Intensity (EUI)

MGM Properties	YTD EUI (kWh/m^2)	% Change Compared to 2019
Macau	21.8	**-12.9%**
Cotai	23.1	**-7.4%**

Table 4. Year to Date Actual Water Use Intensity (WUI)

MGM Properties	YTD WUI (m^3/ m^2)	% Change Compared to 2019
Macau	0.21	**-7.5%**
Cotai	0.14	**-17.4%**

Note: Cotai Area: 374,138 m^2 Macau Area: 211,545 m^2

ISO 50001 Certification Audit

Macau

The ISO 50001 certificate has been received by MGM on 1 April 2021. The certificate will be valid from 3 April 2021 to 2 April 2024.

Cotai

The next external surveillance audit will be scheduled for July 2021.

Target Milestones for Year 2021

The following activities are planned for completion this year:

- To perform ISO 50001 2021 Internal and External Audit for MGM Macau and MGM Cotai
- To perform Carbon Audit 2021 for MGM Macau and MGM Cotai
- To deploy and fine tune DGLux dashboard with heatmap development extension
- To improve reliability and accuracy of MGM Macau LPG metering
- To complete analysis on MGM Cotai spectacle thermal comfort investigation
- To track and monitor changes in air change rates per hour for regular reporting
- To investigate and implement the below list of projects during Year 2021

Table 5. MGM Macau Year 2021 Projects

Item	Initiatives
Electricity	
1	Implement Static Pressure & Supply Air Temperature Set-Point Reset in AHUs (10 units)
2	Smoking Rooms Ventilation Fans Operation Optimization
3	Heat Pump Plant Optimization: Optimize Hot Water Pumps Operation
4	Neon Lighting LED Replacement
5	Implement Motor Efficiency Upgrades (24 units)
6	Chiller Plant Optimization: Chilled Water Differential Pressure Set-point Optimization
7	Implement Lighting & Idle Equipment Timer Control for non-24 Hours Operated Kitchens
8	PICCV Implementation (17 Units)
9	Swimming Pool Heating Optimization
10	Optimize Winter Operation of Cooling Tower Control Strategy - Fan Control
Water	
11	Kitchen Areas - Water Audit and Water Management Practices
12	Retrofit Existing Water Softener serving Podium
13	Optimize Operation of Cooling Tower Control Strategy - Water Conservation

Table 6. MGM Cotai Year 2021 Projects

Item	Initiatives
Electricity	
1	Implement Static Pressure & Supply Air Temperature Set-Point Reset in AHU (16 Units)
2	Smoking Rooms Ventilation Fans Operation Optimization
3	Optimize Hotel FCU Operation with Occupied & Non-Occupied Mode
4	Chiller Plant Optimization: 1. Review Chillers Staging 2. Cooling Tower Fan Speed Optimization 3. Condensing Water Pump Speed Optimization
5	Heat Pump Plant Optimization: Optimize Hot Water Pumps Operation
6	Optimize Hotel Towers Chilled & Hot Water Pumps Operation
7	Chilled Water Differential Pressure Set-Point Reset Logic Implementation
8	Implement Lighting and Idle Equipment Timer Control for non 24 Hours Operated Kitchens
9	Thin Film Solar PV Panel Implementation
10	Replace Insulation on outdoor AHU RA ductwork serving Spectacle area
11	PICCV Implementation (9 Units)
12	Kitchen Exhaust Fan (KEF) Demand Control Ventilation (DCV) and Hood Control for Remaining Kitchen's (e.g. Banquet, Chun, Mansion, Coast.)
13	Relative Humidity Sensors Installation and continuous monitoring for Guest Room & corridors
14	Optimize temperature by adjusting FCU and VAV (flow SPs)
15	Optimize Car Park Ventilation Fans Control
Water	
16	Kitchen Areas - Water Audit and Water Management Practices
17	Extension of CDP back to Cooling Tower Makeup & Floor cleaning

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ACKNOWLEDGEMENT

Energenz has been working with MGM Team as the energy manager to achieve energy savings in MGM properties and feed operational intelligence into the design.

The major goal of this project is not only to realize operation energy savings but also extending the life of equipment, hence reduce CAPEX. MGM has demonstrated continuous commitment to be an environmentally and socially responsible business in the past years. In addition to that Energenz has assisted MGM Macau and Cotai to achieve several other energy management and sustainability awards and certifications, such as ISO 50001 and CEM Energy Savings Award.

Energenz appreciates the great support and trust from MGM Team over the years, especially the Facilities Management and the Sustainability Teams.

1. Portfolio Performance Summary

1.1 Electricity Use Intensity (EUI)

Macau's electricity use intensity (EUI) in April 2021 had decreased by **13.0%** compared to same month in 2019 and decreased by **12.9%** in 2021 compared to 2019. Meanwhile, Cotai's EUI in April 2021 has decreased by **6.8%** compared to same month in 2019 and decreased by **7.4%** in 2021 compared to 2019.

Table 7. Actual Electricity Consumption Intensity (EUI) Comparison

Monthly Period	Macau			Cotai		
	Year 2019 Actual EUI (kWh/m²)	Year 2021 Actual EUI (kWh/m²)	% Change	Year 2019 Actual EUI (kWh/m²)	Year 2021 Actual EUI (kWh/m²)	% Change
January	24.4	20.4	-16.3%	24.5	21.8	-10.7%
February	23.0	19.8	-14.0%	22.9	20.4	-11.0%
March	25.4	23.3	-8.4%	25.0	24.6	-1.6%
April	27.4	23.8	-13.0%	27.3	25.4	-6.8%
May						
June						
July						
August						
September						
October						
November						
December						
Average	**25.1**	**21.8**	**-12.9%**	**24.9**	**23.1**	**-7.4%**

1.2 Water Use Intensity (WUI)

Macau's water use intensity (WUI) in April 2021 decreased by **2.6%** compared to same month in 2019 and decreased by **7.5%** in 2021 compared to 2019. Meanwhile, Cotai's WUI in April 2021 decreased by **10.9%** compared to same month in 2019 and decreased by **17.4%** in 2021 compared to 2019.

Table 8. Actual Water Consumption Intensity (WUI) Comparison

Monthly Period	Macau			Cotai		
	Year 2019 Actual WUI (m^3/m^2)	Year 2021 Actual WUI (m^3/m^2)	% Change	Year 2019 Actual WUI (m^3/m^2)	Year 2021 Actual WUI (m^3/m^2)	% Change
January	0.22	0.19	-13.2%	0.16	0.12	-26.2%
February	0.22	0.19	-11.9%	0.17	0.14	-22.0%
March	0.21	0.20	-2.5%	0.17	0.15	-11.8%
April	0.242	0.236	-2.6%	0.20	0.18	-10.9%
May						
June						
July						
August						
September						
October						
November						
December						
Average	**0.22**	**0.21**	**-7.5%**	**0.17**	**0.14**	**-17.4%**

2. MGM Macau Performance

2.1 Electricity Consumption

The actual electricity consumption and cost in April 2021 decreased by **13.0%** and **13.1%** respectively compared to same month in 2019. To date, MGM Macau has decreased the actual electricity consumption and associated costs by **12.9%** and **17.2%** respectively.

Table 9. Macau - Summary of Actual Monthly Electricity Consumption and Cost

Monthly Period	Actual Electricity Consumption (kWh)			Actual Electricity Cost (MOP)			% Change – CDD
	2019	2021	% Change - Actual	2019	2021	% Change - Actual Cost	
January	5,151,480	4,310,160	-16.3%	6,038,725	4,717,260	-21.9%	-18.5%
February	4,870,560	4,190,760	-14.0%	5,728,930	4,591,170	-19.9%	-3.8%
March	5,381,640	4,931,880	-8.4%	6,311,830	5,387,480	-14.6%	+39.0%
April	5,796,240	5,042,160	-13.0%	6,681,548	5,808,340	-13.1%	-5.5%
May							
June							
July							
August							
September							
October							
November							
December							
Total/Average	**21,199,920**	**18,474,960**	**-12.9%**	**24,761,033**	**20,504,250**	**-17.2%**	**+4.1%**

Important note: *Cooling Degree Days (CDDs) is the main variable taken into consideration on Electricity consumption normalization.*

Energenz summarized the key items which could result in electricity consumption decrease below:

- Outdoor billboards lighting equipment upgrades
- Implement PICCV in AHUs (17 units)
- Implement Static Pressure Set-Point Reset in AHUs (10 units)
- Implement Motor Efficiency Upgrades (24 units)

Table 10 presents the comparison of electricity rates in the year of 2019 and 2021. The electricity rate in April 2021 decreased by **0.1%** compared to same month in 2019.

Table 10. Macau - Summary of Actual Monthly Electricity Rate

Monthly Period	Year 2019 – Electricity Rate (MOP/kWh)	Year 2021 – Electricity Rate (MOP/kWh)	% Change – Electricity Rate
January	1.17	1.09	-6.6%
February	1.18	1.10	-6.9%
March	1.17	1.09	-6.9%
April	1.15	1.15	-0.1%
May			
June			
July			
August			
September			
October			
November			
December			
Total/Average	**1.17**	**1.11**	**-5.1%**

Table 11 summarizes the monthly normalized electricity consumption. The normalized electricity consumption in April 2021 decreased by **12.5%** compared to same month in 2019.

Table 11. Macau - Summary of Normalized Monthly Electricity Consumption

Monthly Period	2019 Normalized Electricity Consumption (kWh)	2021 Actual Electricity Consumption (kWh)	% Change - Normalized
January	4,763,370	4,310,160	-9.5%
February	5,024,953	4,190,760	-16.6%
March	5,398,600	4,931,880	-8.6%
April	5,765,334	5,042,160	-12.5%
May			
June			
July			
August			
September			
October			
November			
December			
Total/Average	**20,952,257**	**18,474,960**	**-11.8%**

2.2 LPG Consumption

LPG consumption can be broken down into three (3) categories: boiler, hotel upper zone DHW and kitchens/restaurants. Since the boiler and restaurants LPG utilization cover over 95% of the overall site LPG consumption, Energenz focuses the analysis on those two areas.

According to Table 12, the actual LPG consumption and cost in April 2021 decreased by **4.6%** and increased by **24.8%** respectively compared to same month in 2019. To date MGM Macau has increased the actual LPG consumption and associated costs by **9.6%** and **40.3%** respectively.

Table 12. Macau - Summary of Actual Monthly LPG Consumption and Cost

Monthly Period	Actual LPG Consumption (kg)			Actual LPG Cost (MOP)			% Change – Min OAT
	2019	2021	% Change - Actual	2019	2021	% Change - Actual Cost	
January	94,330	121,398	+28.7%	706,171	947,708	+34.2%	-29.9%
February	72,191	74,797	+3.6%	415,818	632,778	+52.2%	+10.1%
March	78,807	84,131	+6.8%	453,930	711,751	+56.8%	+6.4%
April	81,430	77,702	-4.6%	522,530	652,357	+24.8%	-2.5%
May							
June							
July							
August							
September							
October							
November							
December							
Total/Average	**326,758**	**358,028**	**+9.6%**	**2,098,449**	**2,944,595**	**+40.3%**	**-2.2%**

Important note: *Minimum Outdoor Air Temperature (OAT), Guest Night and Food Cover are the main variable taken into consideration on LPG consumption normalization.*

Energenz summarized the key items which could result in LPG consumption decrease below:

- Boilers LPG consumption increased by 9.1% in April 2021 while the average OAT was 2.5% less than when compared with the same month in 2019. The LPG consumption in Macau will increase as the building utilizes steam to disinfect the hot water pipe of the heat exchanger. The temperature boost-up is extended until the detection problem is resolved. Currently, there is no end date identified.

- LPG consumption from Kitchen and Restaurants decreased by 3.1% in April 2021 compared with the same month in 2019.
- As previously identified, there is a discrepancy between the total invoice meter and the sum of the LPG submeters. While increase in boiler LPG and decrease in kitchen consumption was noted - this corresponds to a 8.3% difference between actual consumption and totalled sub-meters

Table 13 presents the comparison of LPG rates in the year of 2019 and 2021. The LPG rate in April 2021 increased by **30.8%** compared with the same month in 2019.

Table 13. Macau - Summary of Actual Monthly LPG Rate

Monthly Period	Year 2019 – LPG Rate (MOP/kg)	Year 2021 – LPG Rate (MOP/ kg)	% Change – LPG Rate
January	7.49	7.81	+4.3%
February	5.76	8.46	+46.9%
March	5.76	8.46	+46.9%
April	6.42	8.40	+30.8%
May			
June			
July			
August			
September			
October			
November			
December			
Total/Average	**6.36**	**8.28**	**+30.3%**

Table 14 summarizes the monthly normalized LPG consumption. The normalized consumption in April 2021 increased by **8.3%** compared with the actual consumption in 2021.

Table 14. Macau - Summary of Normalized Monthly LPG Consumption

Monthly Period	2019 Normalized LPG Consumption (kg)	2021 Actual LPG Consumption (kg)	% Change – Normalized
January	116,973	121,398	+3.8%
February	83,731	74,797	-10.7%
March	81,839	84,131	+2.8%
April	71,748	77,702	+8.3%
May			
June			
July			
August			
September			
October			
November			
December			
Total/Average	**354,291**	**358,028**	**+1.1%**

2.3 ISO50001 Energy Consumption Record

The Macau ISO50001 energy performance target is to achieve 13% normalized energy savings by 2025 compared to 2019 baseline. Table 15 below summarizes the site monthly energy consumption (Electricity and LPG combined) compared to 2019 baseline. To date, normalized energy saving achieved up to April 2021 is **9.6%**.

Table 15. Macau ISO 50001 Monthly Energy Consumption

Monthly Period	2019 Normalized Energy Consumption (GJ)	2021 Actual Energy Consumption (GJ)	% Change - Normalized (2021 Vs 2019)
January	22,447	21,016	-6.4%
February	21,883	18,475	-15.6%
March	23,142	21,566	-6.8%
April	24,005	21,672	-9.7%
May			
June			
July			
August			
September			
October			
November			
December			
Total/Average	**91,478**	**82,729**	**-9.6%**

Important note: *Factors used for fuels' unit conversion: 1) Electricity (0.0036) and LPG (0.0453)*

Table 16. Macau ISO 50001 Normalized Monthly Electricity Consumption

Monthly Period	2019 Normalized Electricity Consumption (kWh)	2021 Actual Electricity Consumption (kWh)	% Change - Normalized (2021 Vs 2019)
January	4,763,370	4,310,160	-9.5%
February	5,024,953	4,190,760	-16.6%
March	5,398,600	4,931,880	-8.6%
April	5,765,334	5,042,160	-12.5%
May			
June			
July			
August			
September			
October			
November			
December			
Total/Average	**20,952,257**	**18,474,960**	**-11.8%**

2.4 Water Consumption

Based on the existing water metering system, the overall site water consumption can be broken down into four (4) categories: Kitchen and Restaurants; Hotel/Tower; Fire Services; and Chiller Plant & miscellaneous[2].

According to Table 17, the actual water consumption in April 2021 decreased by **2.6%** compared to same month in 2019. To date, MGM Macau has decreased actual water consumption and associated costs by **7.5%** and **8.6%** respectively.

Table 17. Macau - Summary of Actual Monthly Water Consumption and Cost

Monthly Period	Actual Water Consumption (m^3)			Actual Water Cost (MOP)			Normalization Factors	
	2019	2021	% Change Actual	2019	2021	% Change Actual Cost	% Change – CDD	% Change – Food Covers
January	46,006	39,952	-13.2%	356,970	299,330	-16.1%	-18.5%	-55.0%
February	46,637	41,079	-11.9%	361,510	317,780	-12.1%	-3.8%	-47.7%
March	44,276	43,168	-2.5%	342,840	334,000	-2.6%	+39.0%	-36.4%
April	51,156	49,819	-2.6%	402,280	386,600	-3.9%	-5.5%	-35.8%
May								
June								
July								
August								
September								
October								
November								
December								
Total/ Average	**188,075**	**174,018**	**-7.5%**	**1,463,600**	**1,337,710**	**-8.6%**	**+4.1%**	**-44.0%**

Important note: *Cooling Degree Days (CDDs) and Food Covers are the main variables taken into consideration on Water consumption normalization.*

Energenz summarized the key items which could result in water consumption decrease below:

- Hotel Tower water consumption increased by 1.1% compared to same month in 2019. Moreover, the guest occupancy decreased 7.5% compared to same month in 2019.
- Podium water consumption for April 2021 decreased by 3.3% compared to the same month in 2019. Moreover, the CDD and food cover decreased by 5.5% and 35.8%, respectively, compared to the same month in 2019.

[2] Miscellaneous includes the water consumption from toilets, laundry, etc.

Due to the weather change that occurred on 29 December 2020, the podium FOH and BOH toilet faucet hot water was switched on. Starting 29 December 2020, both podium and tower space heating and the tower domestic hot water will be supported by steam system which is comprised of 1 large boiler and 2 small boilers. As of April 2021, the steam system is still supporting domestic hot water consumption.

Table 18 presents the comparison of water rates in the year of 2019 and 2021. The water rate in April 2021 decreased by **1.3%** compared to same month in 2019. The average utility rate of water has neither significantly increased nor decreased compared with same period in 2019.

Table 18. Macau - Summary of Actual Monthly Water Rates

Monthly Period	Year 2019 – Water Rate (MOP/m^3)	Year 2021 – Water Rate (MOP/m^3)	% Change – Water Rate
January	7.76	7.49	-3.4%
February	7.75	7.74	-0.2%
March	7.74	7.74	-0.1%
April	7.86	7.76	-1.3%
May			
June			
July			
August			
September			
October			
November			
December			
Total/Average	**7.78**	**7.68**	**-1.3%**

Table 19 summarized the monthly normalized water consumption. The normalized water consumption in April 2021 increased by **18.0%** compared to the actual consumption in 2021. Approximately 5% of the increase can be attributed to the higher than expected water consumption from the employee dining room.

Table 19. Macau - Summary of Normalized Monthly Water Consumption

Monthly Period	2019 Normalized Water Consumption (m^3)	2021 Actual Water Consumption (m^3)	% Change - Normalized
January	31,258	39,952	+27.8%
February	34,783	41,079	+18.1%
March	39,846	43,168	+8.3%
April	42,227	49,819	+18.0%
May			
June			
July			
August			
September			
October			
November			
December			
Total/Average	**148,114**	**174,018**	**+17.5%**

2.5 Macau Year 2021 Projects

Energenz will perform analysis and coordinate with MGM to implement the following projects in 2021 and reduce the site energy and water consumption which supporting Energenz annual normalized energy savings target of 2%.

Table 20. MGM Macau Year 2021 Projects

Item	Initiatives
Electricity	
1	Implement Static Pressure & Supply Air Temperature Set-Point Reset in AHU (10 units)
2	Smoking Rooms Ventilation Fans Operation Optimization
3	Heat Pump Plant Optimization: Optimize Hot Water Pumps Operation
4	Neon Lighting LED Replacement
5	Implement Motor Efficiency Upgrades (24 units)
6	Chiller Plant Optimization: Chilled Water Differential Pressure Set-point Optimization
7	Implement Lighting & Idle Equipment Timer Control for non-24 Hours Operated Kitchens
8	PICCV Implementation (17 Units)
9	Swimming Pool Heating Optimization
10	Optimize Winter Operation of Cooling Tower Control Strategy - Fan Control
Water	
11	Kitchen Areas - Water Audit and Water Management Practices
12	Retrofit Existing Water Softener serving Podium
13	Optimize Operation of Cooling Tower Control Strategy - Water Conservation

3. MGM Cotai Performance

3.1 Electricity

The actual electricity consumption and cost in April 2021 decreased by **6.8%** and **6.9%** respectively compared to the same month in 2019. To date, MGM Cotai has decreased actual electricity consumption and associated costs by **6.8%** and **11.3%** respectively.

Table 21. Cotai - Summary of Actual Monthly Electricity Consumption and Cost

Monthly Period	Actual Electricity Consumption (kWh)			Actual Electricity Cost (MOP)			% Change – CDD
	2019	2021	% Change - Actual	2019	2021	% Change - Actual Cost	
January	8,938,560	8,168,400	-8.6%	10,919,060	9,315,910	-14.7%	-18.5%
February	8,560,200	7,637,640	-10.8%	10,478,873	8,737,740	-16.6%	-3.8%
March	9,350,520	9,197,160	-1.6%	11,429,480	10,499,490	-8.1%	+39.0%
April	10,207,920	9,517,560	-6.8%	12,272,805	11,430,660	-6.9%	-5.5%
May							
June							
July							
August							
September							
October							
November							
December							
Total/Average	**37,057,200**	**34,520,760**	**-6.8%**	**45,100,218**	**39,983,800**	**-11.3%**	**+4.1%**

***Important note**: Cooling Degree Days (CDDs) is the main variable taken into consideration on Electricity consumption normalization. The e-Bus electricity consumption has been included with the presented electricity consumption above.*

Table 22 presents the monthly e-bus electricity consumption in MGM Cotai. The e-bus electricity consumption in April 2021 increased by **3.5%** compared to same month in 2019. Please note that consumption related to the e-bus will continue to increase as compared to 2019 as the E-bus first month of operation started in January 2019.

Table 22. Cotai - Summary of Monthly e-bus Electricity Consumption

Monthly Period	E-bus Electricity Consumption (kWh)		
	2019	2021	% Change - Actual
January	28,305	44,839	+58.4%
February	24,007	32,534	+35.5%
March	34,315	34,902	+1.7%
April	36,147	37,412	+3.5%
May			
June			
July			
August			
September			
October			
November			
December			
Total/Average	**122,773**	**149,687**	**+21.9%**

Table 23 presents the comparison of electricity rates in the year of 2019 and 2021. The electricity rate in April 2021 decreased by **0.1%** compared to same month in 2019.

Table 23. Cotai - Summary of Actual Monthly Electricity Rates

Monthly Period	Year 2019 – Electricity Rate (MOP/kWh)	Year 2021 – Electricity Rate (MOP/kWh)	% Change – Electricity Rate
January	1.22	1.14	-6.6%
February	1.22	1.14	-6.5%
March	1.22	1.14	-6.6%
April	1.20	1.20	-0.1%
May			
June			
July			
August			
September			
October			
November			
December			
Total/Average	**1.22**	**1.16**	**-5.0%**

Table 24 summarized the monthly normalized electricity consumption. The normalized electricity consumption in April 2021 decreased by **5.9%** compared to same month in 2019.

Table 24. Cotai - Summary of Normalized Monthly Electricity Consumption

Monthly Period	2019 Normalized Electricity Consumption (kWh)	2021 Actual Electricity Consumption (kWh)	% Change - Normalized
January	8,594,286	8,168,400	-5.0%
February	8,991,397	7,637,640	-15.1%
March	9,558,635	9,197,160	-3.8%
April	10,115,378	9,517,560	-5.9%
May			
June			
July			
August			
September			
October			
November			
December			
Total/Average	**37,259,696**	**34,520,760**	**-7.4%**

3.2 Natural Gas Consumption

The actual natural gas consumption and cost in April 2021 decreased by **19.0%** and **18.7%** respectively compared to same month in 2019. To date, MGM Cotai has decreased actual natural gas consumption and associated costs by **37.6%** and **37.2%** respectively.

Table 25. Cotai - Summary of Actual Monthly Natural Gas Consumption and Cost

Monthly Period	Actual Natural Gas Consumption (m^3)			Actual Natural Gas Cost (MOP)			% Change – Food Covers
	2019	2021	% Change - Actual	2019	2021	% Change - Actual Cost	
January	112,881	60,643	-46.3%	671,460	363,990	-45.8%	-53.9%
February	99,721	54,221	-45.6%	594,320	325,930	-45.2%	-46.8%
March	112,506	68,066	-39.5%	669,261	407,980	-39.0%	-42.2%
April	107,194	86,866	-19.0%	638,120	518,960	-18.7%	-42.5%
May							
June							
July							
August							
September							
October							
November							
December							
Total/Average	**432,302**	**269,796**	**-37.6%**	**2,573,161**	**1,616,860**	**-37.2%**	**-46.3%**

Important note: *Food cover is the main variable taken into consideration on natural gas consumption normalization.*

Energenz summarized the key items which could result in natural gas consumption decrease below:

- Boiler Natural Gas consumption decreased by 7.0% in April 2021 compared to same month in 2019. Natural gas consumption will increase to accommodate the disinfecting procedure. The temperature boost-up is extended until the detection problem is resolved. Currently, there is no end date identified.

- Kitchen and Restaurants natural gas consumption, which constitutes to around 85 - 99% of the overall site consumption, decreased by 17.8% in April 2021 compared to same month in 2019, while food covers had decreased by 42.5%.

Table 26 presents the comparison of natural gas rates in the year of 2019 and 2021. The natural gas rate in April 2021 increased by **0.4%** compared with the same month in 2019.

Table 26. Cotai - Summary of Actual Monthly Natural Gas Rates

Monthly Period	Year 2019 – Natural Gas Rate (MOP/m³)	Year 2021 – Natural Gas Rate (MOP/m³)	% Change – Natural Gas Rate
January	5.95	6.00	+0.9%
February	5.96	6.01	+0.9%
March	5.95	5.99	+0.8%
April	5.95	5.97	+0.4%
May			
June			
July			
August			
September			
October			
November			
December			
Total/Average	**5.95**	**6.00**	**+0.7%**

3.3 ISO50001 Energy Consumption Record

Cotai ISO50001 energy target is to achieve 13% normalized energy consumption reduction by 2025 compared to 2019 baseline. Based on the official agreement with MGM Cotai, e-bus consumption should be included for the energy consumption tracking as part of the ISO50001 requirement. Table 27 below summarizes the site monthly energy consumption (Electricity and natural gas combined) compared to 2019 baseline. The year-to-date normalized energy saving achieved in April 2021 is **11.5%.**

Table 27. Cotai ISO 50001 Normalized Monthly Energy Consumption

Monthly Period	2019 Normalized Energy Consumption (GJ)	2021 Actual Energy Consumption (GJ)	% Change - Normalized (2021 Vs 2019)
January	35,529	31,511	-11.3%
February	36,539	29,377	-19.6%
March	38,505	35,472	-7.9%
April	40,368	37,277	-7.7%
May			
June			
July			
August			
September			
October			
November			
December			
Total/Average	**150,940**	**133,637**	**-11.5%**

***Important note**: Factors used for fuels' unit conversion: 1) Electricity (0.0036) and Natural Gas (0.0347), e-bus consumption is included*

Table 28. Cotai ISO 50001 Normalized Monthly Electricity Consumption

Monthly Period	2019 Normalized Electricity Consumption (kWh)	2021 Actual Electricity Consumption (kWh)	% Change - Normalized (2021 Vs 2019)
January	8,594,286	8,168,400	-5.0%
February	8,991,397	7,637,640	-15.1%
March	9,558,635	9,197,160	-3.8%
April	10,115,378	9,517,560	-5.9%
May			
June			
July			
August			
September			
October			
November			
December			
Total/Average	**37,259,696**	**34,520,760**	**-7.4%**

***Important note**: e-bus consumption is included*

3.4 Water Consumption

According to Table 29, the actual water consumption in April 2021 decreased by **10.9%** compared to same month in 2019. To date, MGM Cotai has decreased actual water consumption and associated costs by **16.9%** and **16.8%** respectively.

Table 29. Cotai - Summary of Actual Monthly Water Consumption and Cost

Monthly Period	Actual Water Consumption (m^3)			Actual Water Cost (MOP)			Normalization Factors	
	2019	2021	% Change Actual	2019	2021	% Change Actual Cost	% Change – CDD	% Change – Food Covers
January	57,331	43,295	-24.5%	445,450	336,678	-24.4%	-18.5%	-53.9%
February	65,011	50,847	-21.8%	505,050	395,188	-21.8%	-3.8%	-46.8%
March	63,136	55,680	-11.8%	488,590	432,418	-11.5%	+39.0%	-42.2%
April	73,577	65,523	-10.9%	571,320	509,018	-10.9%	-5.5%	-42.5%
May								
June								
July								
August								
September								
October								
November								
December								
Total/ Average	**259,055**	**215,345**	**-16.9%**	**2,010,410**	**1,673,301**	**-16.8%**	**+4.1%**	**-46.3%**

***Important note**: Cooling Degree Days (CDDs) and Food Covers are the main variables taken into consideration on Water consumption normalization.*

Energenz summarized the key items which could result in water consumption decrease below:

- Cooling Tower Make Up water consumption decreased by 4.7% in April 2021 compared to same month in 2019. At the same time, CDD decreased by 5.5% in April 2021 compared to same month in 2019.
- Hotel Tower & Podium water consumption decreased by 17.1% in April 2021 compared to same month in 2019. Moreover, guest occupancy decreased 29.6% in April 2021 compared to same month in 2019.

Table 30 presents the comparison of water rates in the year of 2019 and 2021. The water rate in April 2021 increased by **0.05%** compared to same month in 2019. The average utility rate of water has neither significantly increased nor decreased compared with same period in 2019.

Table 30. Cotai - Summary of Actual Monthly Water Rates

Monthly Period	Year 2019 – Water Rate (MOP/m³)	Year 2021 – Water Rate (MOP/m³)	% Change – Water Rate
January	7.77	7.78	+0.1%
February	7.77	7.77	+0.04%
March	7.74	7.77	+0.4%
April	7.76	7.77	+0.05%
May			
June			
July			
August			
September			
October			
November			
December			
Total/Average	**7.76**	**7.77**	**+0.1%**

Table 31 summarized the monthly normalized water consumption. The normalized water consumption in April 2021 increased by **11.7%** compared to the actual consumption in 2021.

Table 31. Cotai - Summary of Normalized Monthly Water Consumption

Monthly Period	2019 Normalized Water Consumption (m³)	2021 Actual Water Consumption (m³)	% Change - Normalized
January	45,293	43,295	-4.4%
February	48,765	50,847	+4.3%
March	55,357	55,680	+0.6%
April	58,684	65,523	+11.7%
May			
June			
July			
August			
September			
October			
November			
December			
Total/Average	**208,098**	**215,345**	**+3.5%**

3.5 Cotai Year 2021 Projects

Energenz will perform analysis and coordinate with MGM to implement the following projects in 2021 and reduce the site energy and water consumption which supporting Energenz annual normalized energy savings target of 2%.

Table 32. MGM Cotai Year 2021 Projects

Item	Initiatives
Electricity	
1	Implement Static Pressure & Supply Air Temperature Set-Point Reset in AHU (16 Units)
2	Smoking Rooms Ventilation Fans Operation Optimization
3	Optimize Hotel FCU Operation with Occupied & Non-Occupied Mode
4	Chiller Plant Optimization: 1. Review Chillers Staging 2. Cooling Tower Fan Speed Optimization 3. Condensing Water Pump Speed Optimization
5	Heat Pump Plant Optimization: Optimize Hot Water Pumps Operation
6	Optimize Hotel Towers Chilled & Hot Water Pumps Operation
7	Chilled Water Differential Pressure Set-Point Reset Logic Implementation
8	Implement Lighting and Idle Equipment Timer Control for non 24 Hours Operated Kitchens
9	Thin Film Solar PV Panel Implementation
10	Replace Insulation on outdoor AHU RA ductwork serving Spectacle area
11	PICCV Implementation (9 Units)
12	Kitchen Exhaust Fan (KEF) Demand Control Ventilation (DCV) and Hood Control for Remaining Kitchen's (e.g. Banquet, Chun, Mansion, Coast.)
13	Relative Humidity Sensors Installation and continuous monitoring for Guest Room & corridors
14	Optimize temperature by adjusting FCU and VAV (flow SPs)
15	Optimize Car Park Ventilation Fans Control
Water	
16	Kitchen Areas - Water Audit and Water Management Practices
17	Extension of CDP back to Cooling Tower Makeup & Floor cleaning

Appendix A

Regression Analysis

The 2019 normalized consumprion modelling is remained used for tracking the agreed normalized energy savings target of 4% accurately for MGM Macau and MGM Cotai. The following tables show a summary of the selected independent variables for each utility which are used for the normalized consumption modelling in 2021.

MGM Macau

Utilities	2021	
Electricity (kWh)	CDD (Base T=17°C)	
LPG (kg)	Min OAT	
Water (m^3)	CDD (Base T=17°C)	Food Cover

MGM Cotai

Utilities	2021	
Electricity (kWh)	CDD (Base T=17°C)	
Natural Gas (m^3)	Min OAT	
Water (m^3)	CDD (Base T=17°C)	Food Cover

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